Phase1_analysis

March 9, 2026

[70]:
```
import importlib
import pandas as pd
import numpy as np
import re
from pathlib import Path

import functions
importlib.reload(functions)
from functions import *

df_1111 = pd.read_csv("AI_Lit_Que_1111.csv")
df_1204 = pd.read_csv("AI_Lit_Que_1204.csv")

# add source/course label
df_1111["course"] = "1111"
df_1204["course"] = "1204"

print(df_1111.shape, df_1204.shape)
```

```
(118, 37) (23, 37)
```

[71]:
```
# rename_map is defined in functions.py
df_1111 = df_1111.rename(columns=rename_map)
df_1204 = df_1204.rename(columns=rename_map)

df = pd.concat([df_1111, df_1204], ignore_index=True)
```

1 Preprocessing

[72]:
```
# clean_text, extract_leading_code, cronbach_alpha, add_combined_panel
# are defined in functions.py

df = df.applymap(clean_text)

# columns that should NOT go through leading-number extraction
exclude_from_numeric = {
    "id",
```

```
    "course",
    "ses_lang_cantonese",
    "ses_lang_mandarin",
    "ses_lang_english",
    "ses_lang_other",
}

special_attention_cols = {
    "ses_school_type",        # mostly nominal
    "ses_housing_type",       # not perfectly ordinal
    "ses_household_income",   # contains "Prefer not to say"
}

# extract from all remaining renamed questionnaire columns
numeric_source_cols = [
    col for col in df.columns
    if col not in exclude_from_numeric
]
for col in numeric_source_cols:
    df[col + "_num"] = df[col].apply(extract_leading_code)

# language columns should stay as binary indicators
language_cols = [
    "ses_lang_cantonese",
    "ses_lang_mandarin",
    "ses_lang_english",
    "ses_lang_other",
]
for col in language_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")
```

---

2 EDA

2.1 Descriptive statistics

2.1.1 Mean comparison

SES

- Class 1111 is very slightly below the pooled SES average
- Class 1204 is slightly above the pooled SES average

The difference is present but not large in raw standardized units.

- *Skew* < 0: SES is somewhat left-skewed, especially in 1204. More observations are clustered on the higher side, with a tail toward lower SES-index values.

AI Literacy

- Class 1204 appears to have higher AI literacy on average than class 1111. The gap here is much more visible than for SES.

[73]:
```
df, meta = prepare_dataset("AI_Lit_Que_1111.csv", "AI_Lit_Que_1204.csv")

construct_summary_table(df)
```

[73]:
```
     sample          construct    n          mean        sd    median  \
0      1111          SES index  118 -2.184274e-02  0.570420  0.003448
1      1111  AI literacy score  118  3.490678e+00  0.409291  3.500000
2      1204          SES index   23  1.120627e-01  0.569433  0.235097
3      1204  AI literacy score   23  3.800000e+00  0.450252  3.900000
4  Combined          SES index  141 -2.834612e-17  0.570388  0.020992
5  Combined  AI literacy score  141  3.541135e+00  0.430128  3.500000

        iqr      skew
0  0.792470 -0.409507
1  0.500000  0.266702
2  0.661270 -0.804122
3  0.750000  0.134169
4  0.801864 -0.460321
5  0.600000  0.294079
```

2.1.2 Measurement consistency

SES

- SES shows **moderate** consistency across both classes (pooled Cronbach's = 0.69).

AI Literacy

- AI literacy shows **low** consistency across both classes (pooled Cronbach's = 0.52; even lower in 1111).

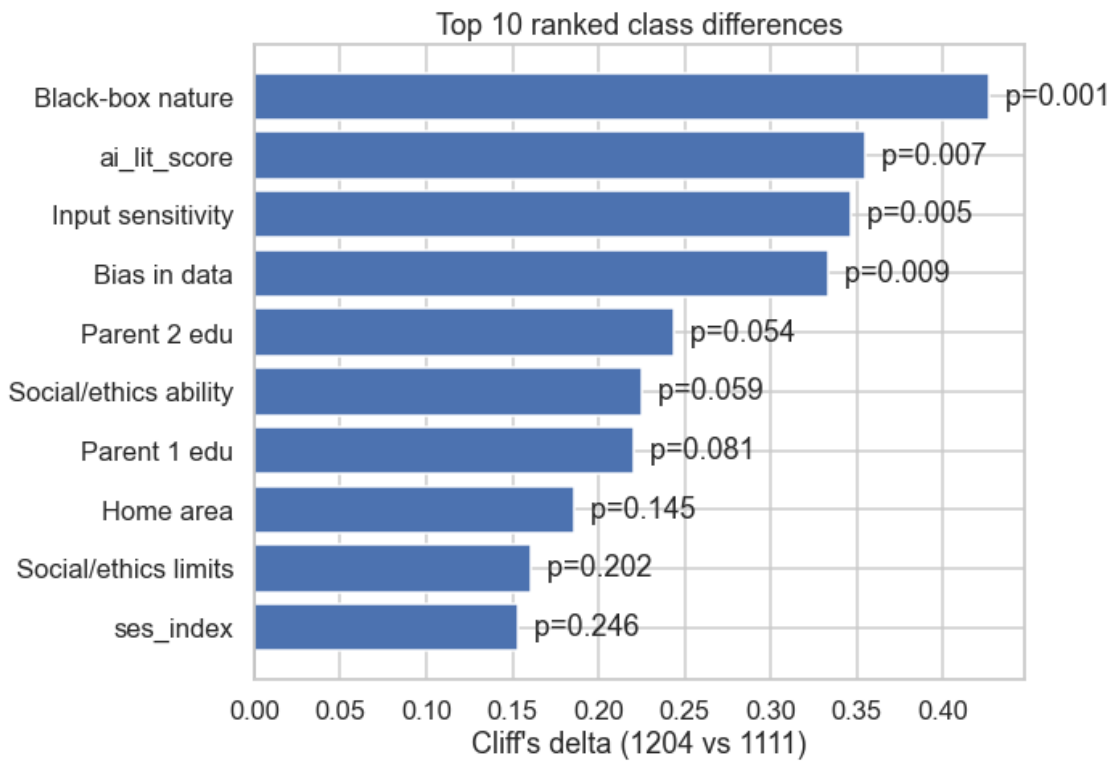
The AI literacy items do not appear to function as a highly homogeneous single scale in this sample.

[74]:
```
reliability_table(df, meta)
```

[74]:
```
     sample              construct     alpha  n_complete
0      1111  SES index ingredients  0.689986         118
1      1111      AI literacy score  0.466021         118
2      1204  SES index ingredients  0.739366          23
3      1204      AI literacy score  0.624770          23
4  Combined  SES index ingredients  0.693905         141
5  Combined      AI literacy score  0.523376         141
```

2.1.3 Effect sizes: Two-sided Mann–Whitney U test

[75]:
```
eff = effect_size_table(df, meta)
plot_ranked_effect_sizes(eff, top_n=10)
```



- Class 1204 tends to score higher than class 1111 on several AI literacy components, especially the more conceptual items.
- By contrast, the SES differences are weaker and mostly not statistically convincing.

2.2 Visualizations

2.2.1 Item profiles

[76]:
```
plot_item_profile(df, meta, construct="SES")
```

[77]:
```
plot_item_profile(df, meta, construct="AI")
```

2.2.2 Distribution of composite scores

[78]:
```
plot_ses_ai_correlation_heatmaps(df, meta)
```

- In the pooled sample, most correlations are weak to modest, with a tendency toward small positive associations rather than a single dominant pattern.
- The 1204 panel shows more extreme values, but given the much smaller subgroup size, those correlations are likely less stable.

---

3 SES Effect on AI Literacy (No mediation)

$$\text{AI literacy} = \beta_0 + \beta_1 \text{SES} + \varepsilon$$

[79]:
```
def run_total_effect_models(df):
    results = []

    for sample in ["1111", "1204", "Combined"]:
        res, _, _ = fit_total_effect_model(df, sample=sample)
        results.append(res)

    return pd.concat(results, ignore_index=True)

run_total_effect_models(df)
```

[79]:
```
      sample    n  beta_ses_std    se_hc3     t_hc3     p_hc3  ci_low_95  \
0       1111  118      0.194558  0.084811  2.294033  0.021789   0.028333
1       1204   23      0.109333  0.270676  0.403926  0.686267  -0.421182
2   Combined  141      0.195893  0.076050  2.575846  0.010000   0.046838

   ci_high_95  r_squared  spearman_rho  spearman_p
0    0.360784   0.037853      0.199891    0.029991
1    0.639848   0.011954      0.013161    0.952473
```

```
2    0.344948    0.038374      0.201231     0.016723
```

- In the **combined sample**, SES shows a small but statistically supported positive association with AI literacy. Students with higher SES tend to have slightly higher AI literacy scores.
- In **class 1111**, the estimated SES effect is again positive and modest, and the result is statistically significant.
- In **class 1204**, the estimated effect is also positive, but highly uncertain and not statistically supported — most likely due to the much smaller sample size rather than evidence that no relationship exists.

The effect size is not large. In both 1111 and the combined sample, a one standard deviation increase in SES is associated with only about a 0.2 SD increase in AI literacy.

The explained variance is low (R^2 0.04), meaning SES accounts for only a small part of the variation in AI literacy. SES appears to matter, but is clearly not the only factor.

- The Spearman correlations are also positive in 1111 and in the combined sample, supporting the same conclusion from a rank-based perspective.

3.1 Which SES dimensions matter most?

[80]:
```
ai_conceptual_items = [
    "ai_concept_data_bias_scored_num",
    "ai_concept_blackbox_scored_num",
    "ai_concept_input_variation_scored_num",
    "ai_concept_prompt_wording_scored_num",
    "ai_concept_social_ethics_scored_num",
]

ai_ability_items = [
    "ai_ability_training_data_scored_num",
    "ai_ability_explainability_scored_num",
    "ai_ability_input_sensitivity_scored_num",
    "ai_ability_prompting_scored_num",
    "ai_ability_social_ethics_scored_num",
]

def add_ai_subscales(df):
    d = df.copy()

    d["ai_conceptual_score"] = d[ai_conceptual_items].mean(axis=1)
    d["ai_ability_score"] = d[ai_ability_items].mean(axis=1)

    return d

df_domain = add_ai_subscales(df)
domain_results, domain_models = run_domain_models(df_domain)
domain_results.round(3)
```

[80]:

	sample	outcome	term	beta_std	\
0	1111	AI conceptual understanding	Home area	0.121	
1	1111	AI conceptual understanding	Device access	0.060	
2	1111	AI conceptual understanding	Parent 2 education	0.070	
3	1111	AI conceptual understanding	Parent 1 education	-0.054	
4	1111	AI conceptual understanding	Internet quality	0.026	
5	1111	AI conceptual understanding	Household income	-0.003	
6	1111	AI conceptual understanding	Low financial strain	0.002	
7	1111	AI ability/confidence	Internet quality	-0.200	
8	1111	AI ability/confidence	Household income	0.212	
9	1111	AI ability/confidence	Device access	0.146	
10	1111	AI ability/confidence	Parent 1 education	-0.176	
11	1111	AI ability/confidence	Home area	0.130	
12	1111	AI ability/confidence	Parent 2 education	0.088	
13	1111	AI ability/confidence	Low financial strain	0.015	
14	1204	AI conceptual understanding	Low financial strain	0.481	
15	1204	AI conceptual understanding	Parent 2 education	0.747	
16	1204	AI conceptual understanding	Device access	-0.181	
17	1204	AI conceptual understanding	Parent 1 education	-0.417	
18	1204	AI conceptual understanding	Household income	-0.283	
19	1204	AI conceptual understanding	Internet quality	-0.200	
20	1204	AI conceptual understanding	Home area	-0.262	
21	1204	AI ability/confidence	Household income	0.585	
22	1204	AI ability/confidence	Parent 2 education	0.481	
23	1204	AI ability/confidence	Home area	-0.466	
24	1204	AI ability/confidence	Device access	0.172	
25	1204	AI ability/confidence	Internet quality	-0.087	
26	1204	AI ability/confidence	Parent 1 education	-0.122	
27	1204	AI ability/confidence	Low financial strain	0.078	
28	Combined	AI conceptual understanding	Home area	0.156	
29	Combined	AI conceptual understanding	Household income	-0.133	
30	Combined	AI conceptual understanding	Parent 2 education	0.148	
31	Combined	AI conceptual understanding	Device access	0.049	
32	Combined	AI conceptual understanding	Parent 1 education	-0.057	
33	Combined	AI conceptual understanding	Low financial strain	0.034	
34	Combined	AI conceptual understanding	Internet quality	-0.024	
35	Combined	AI ability/confidence	Household income	0.234	
36	Combined	AI ability/confidence	Internet quality	-0.195	
37	Combined	AI ability/confidence	Device access	0.153	
38	Combined	AI ability/confidence	Parent 1 education	-0.153	
39	Combined	AI ability/confidence	Home area	0.116	
40	Combined	AI ability/confidence	Parent 2 education	0.096	
41	Combined	AI ability/confidence	Low financial strain	0.012	

	se_hc3	t_hc3	p_hc3	ci_low_95	ci_high_95	r_squared	n
0	0.121	1.001	0.317	-0.116	0.357	0.024	118
1	0.086	0.694	0.488	-0.109	0.228		

2	0.146	0.479	0.632	-0.216	0.356		
3	0.141	-0.384	0.701	-0.331	0.223		
4	0.104	0.255	0.799	-0.177	0.230		
5	0.111	-0.030	0.976	-0.222	0.215		
6	0.141	0.011	0.991	-0.275	0.278		
7	0.093	-2.140	0.032	-0.383	-0.017	0.121	118
8	0.109	1.944	0.052	-0.002	0.425		
9	0.078	1.873	0.061	-0.007	0.298		
10	0.172	-1.020	0.308	-0.513	0.162		
11	0.130	1.001	0.317	-0.125	0.384		
12	0.196	0.446	0.656	-0.297	0.473		
13	0.198	0.075	0.940	-0.373	0.403		
14	0.230	2.091	0.037	0.030	0.932	0.343	23
15	0.498	1.501	0.133	-0.229	1.724		
16	0.134	-1.346	0.178	-0.445	0.082		
17	0.352	-1.187	0.235	-1.106	0.272		
18	0.394	-0.717	0.474	-1.055	0.490		
19	0.330	-0.605	0.545	-0.848	0.448		
20	0.691	-0.379	0.705	-1.615	1.092		
21	0.325	1.797	0.072	-0.053	1.222	0.335	23
22	0.418	1.151	0.250	-0.338	1.300		
23	0.429	-1.086	0.277	-1.306	0.374		
24	0.373	0.461	0.645	-0.559	0.903		
25	0.306	-0.284	0.776	-0.686	0.512		
26	0.433	-0.281	0.779	-0.969	0.726		
27	0.360	0.217	0.829	-0.628	0.784		
28	0.110	1.424	0.154	-0.059	0.371	0.046	141
29	0.104	-1.275	0.202	-0.338	0.072		
30	0.125	1.188	0.235	-0.096	0.393		
31	0.084	0.583	0.560	-0.116	0.214		
32	0.119	-0.475	0.635	-0.290	0.177		
33	0.121	0.279	0.780	-0.203	0.271		
34	0.090	-0.261	0.794	-0.201	0.154		
35	0.098	2.396	0.017	0.043	0.426	0.13	141
36	0.087	-2.241	0.025	-0.366	-0.025		
37	0.069	2.202	0.028	0.017	0.289		
38	0.150	-1.024	0.306	-0.446	0.140		
39	0.116	1.001	0.317	-0.111	0.343		
40	0.170	0.562	0.574	-0.238	0.429		
41	0.170	0.071	0.943	-0.320	0.345		

- For **class 1111**, the SES predictors explained very little variance in conceptual AI literacy ($R^2 = 0.024$), and no SES component showed a clear association with conceptual understanding.
  - Household income and device access showed positive near-significant associations.
  - Internet quality showed a statistically significant negative coefficient.
- For **class 1204**, both models had relatively large R^2 values (0.343 for conceptual understanding and 0.335 for ability/confidence). Most coefficients had wide confidence intervals and were not statistically supported.

  - The one notable result was a positive coefficient for low financial strain in the conceptual model, suggesting fewer financial constraints may be associated with better conceptual AI understanding in this class.
- In the **combined sample**, the pattern is clearer and more stable. The SES predictors explained little variance in conceptual AI literacy ($R^2 = 0.046$), and no SES component emerged as a clear predictor of conceptual understanding.

[ ]:

```
```